



Radiant QuikQA

User's Guide

Version 1.0.1 · Powered by PyLinac

A PyLinac-powered medical physics QA toolkit
for routine linac QA, SBRT QA, and TG-51 reference dosimetry

Radiant Medical Physics Consulting LLC

Todd Bernath, MS, DABR

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1. Introduction

Radiant QuikQA is a focused, fast medical-physics QA tool built on top of the PyLinac analysis library, with a native TG-51 reference-dosimetry engine added in the v0.8.x and v0.9.x series. It is designed for the working physicist who needs to run routine and SRS/SBRT linac QA quickly, perform monthly/annual reference dosimetry, produce branded PDF reports for the clinical record, and have results available both at a desktop GUI and through Eclipse's published-scripts surface via an ESAPI launcher.

This guide covers version 1.0.1, which adds parallel-plate chamber cross-calibration (TG-51 / TG-39) and the calibration-expiry warning system on top of the substantial work shipped in v0.5 through v1.0.0. It is written for physicists who will use the software clinically. Familiarity with PyLinac's clinical conventions (TG-142, AAPM MPPG) and AAPM TG-51 / Report 374 is assumed; this guide focuses on what each tab does, what inputs it expects, what outputs it produces, and the configuration system that ties everything together.

1.1 What Radiant QuikQA is

- A standalone Windows desktop application that wraps PyLinac for routine, repeatable linac QA, with native engines for TG-51 reference dosimetry layered on top.
- A multi-clinic configuration system: machines, clinics, per-machine tolerances, per-machine default folders, per-machine beam catalogs, and per-clinic default chamber and electrometer settings are all stored centrally and persist across sessions.
- A branded PDF report generator that emits clinical reports stamped with the Radiant logo and machine / protocol / chamber metadata.
- A command-line interface (CLI) callable from scripts, scheduled tasks, and the bundled Eclipse ESAPI launcher. The CLI covers Winston-Lutz, Picket Fence, VMAT, CT, Portal Dosimetry, TG-51 photons, TG-51 electrons, and TG-39 cross-cal.
- Open to inspection: every PyLinac module is a thin wrapper around a documented PyLinac entry point. The TG-51 engines are native to Radiant QuikQA; their math is verified against AAPM TG-51 (1999), the 2014 Addendum, AAPM Report 374 (2022), and Burns/Ding/Rogers (1996).

1.2 What Radiant QuikQA is not

- It is not a replacement for vendor commissioning or acceptance testing. It is a routine-QA and reference-dosimetry tool.
- It is not a treatment planning system, dose calculation engine, or DICOM viewer. Patient data has no place in it.
- It is not certified medical device software. The clinical user is responsible for validating each module against vendor or independent reference values before using the results clinically.

About the BETA marker

Modules carrying a BETA label have completed software testing but have not yet been validated against independently measured reference values at the user's site. Use BETA modules in parallel with your current vendor analysis until you have your own validation data; do not use them as the sole QA basis for clinical decisions.

v1.0.1 BETA modules: CT/CBCT Phantom QA, and the 3D Isocenter (Hancock) variant of the Starshot tab.

1.3 TG-51 chambers and validation

Every chamber and electrometer in the TG-51 catalog ships with a status of UNVALIDATED. The math in the engine is independent of validation status; the status flag indicates whether your clinic has confirmed agreement with an independent calculation method (RadCalc, spreadsheet, manual hand calc) on a real cal. Once that agreement is confirmed within 0.5%, edit the chamber's status to VALIDATED in `radiant_core/dosimetry/tg51_chambers_electrons.py` (or `_photons.py`) with a notes entry. Until then, treat the engine output as a check, not the source of truth.

PTW 30013 specifically: the v0.9.0 release uses $k_{R50,ecal} = 0.897$ (from TG-51 Table III N30001 cross-walked through the 2014 Addendum). Some clinic spreadsheets have used 0.898 (a historical transcription confusion with the N31003 row). The 0.1% difference is within Table III's quoted precision; do not be surprised if your first electron cal disagrees with your spreadsheet by a fraction of a percent on this chamber.

1.4 Conventions used in this guide

Items shown in monospaced font are file names, file paths, command-line text, or values typed in literally.

Yellow note boxes call out non-obvious behavior worth noting before clinical use.

Red-edged warning boxes call out conditions that can produce silently-wrong results or break the workflow.

2. System Requirements and Installation

2.1 Supported environment

Item	Requirement
OS	Windows 10 or Windows 11, 64-bit
Python	Python 3.12.x (only required if running from source)
PyLinac	3.43.2 (pinned; do not change without re-validating)
.NET	4.8 (required by the Eclipse launcher only)
Eclipse	v15.6 (for the ESAPI launcher path; not required for standalone GUI)
Inno Setup	6.7.1 (build environment only; not required to run)
Disk	~300 MB after install; reports folder grows with use
RAM	4 GB minimum; 8 GB recommended for large CBCT stacks

2.2 Installation

Three install modes are supported:

Mode A: Production install (Inno Setup .exe)

On a clinical workstation, install via the bundled Inno Setup installer. By default it installs to:

```
C:\Radiant_QuikQA\
```

The installer adds a Start menu entry and a desktop shortcut that launches the GUI. After installation:

1. Add the install folder to Norton AntiVirus's exclusion list if your endpoint protection blocks unsigned packaged Python binaries. Exclude the folder, not individual files.
2. Launch the GUI once and walk through Setup → Clinics... and Setup → Machines... to confirm the default fleet is correct for your site (see Section 4).
3. Configure clinic-level chamber and electrometer defaults if you plan to use TG-51. See Section 4.2.

Mode B: Pre-built distribution zip

For colleagues or secondary workstations where running the Inno installer is inconvenient, you can ship a zip of the pre-built dist/ folder (produced by build_all.ps1). The recipient extracts to any folder and runs RadiantQuikQA.exe directly. No Python required. Configuration bootstraps under %LOCALAPPDATA% on first launch.

Mode C: Development install (Python from source)

For development or in-house customization, install Python 3.12, then from the source tree:

```
cd C:\Radiant_QuikQA\radiant_quikqa
```

```
python -m pip install -e .
python -m pip install pytest
python -m pytest -q
python -m radiant_gui.app
```

Pinned dependencies are declared in pyproject.toml. Do not upgrade PyLinac without rerunning the test suite and re-validating clinical modules.

2.3 Where Radiant QuikQA stores its data

Radiant QuikQA never touches the user's Documents folder, because Documents is often redirected to OneDrive on managed Windows installs and the redirect can fail under load. All Radiant QuikQA state lives under:

%LOCALAPPDATA%\RadiantQuikQA\

Inside that folder, you will find:

File / Folder	Purpose
machines.json	The machine fleet (display names, MLC types, tolerances, default folders, and the per-machine beam catalog added in v0.9.2)
locations.json	Clinic locations, in-service machines, and (since v0.9.2) per-clinic default chamber + electrometer settings
settings.json	Per-workstation preferences (reports folder root, last-used paths, stamping toggles)
Reports\	PDF report output, organized into per-machine subfolders by default
tg51_history\	Per-(machine, beam) TG-51 photon calibration history JSON files (v0.8.0+)
tg51_electron_history\	Per-(machine, beam) TG-51 electron calibration history JSON files (v0.9.0+)
tg51_cross_cal_history\	Per-(PP chamber, serial) cross-cal records (v1.0.1+)

Backing up your configuration

The JSON files above contain all of your machine tolerances, clinic settings, beam catalogs, and TG-51 history. Back them up periodically; if you reinstall or move workstations, restoring these files restores your configuration.

Reports are saved to %LOCALAPPDATA%\RadiantQuikQA\Reports\<machine>\ unless you change the reports root in Preferences.

3. Application Overview

3.1 Launching the GUI

Production install: double-click the Radiant QuikQA shortcut on the desktop, or launch from the Start menu.

From source:

```
cd C:\Radiant_QuikQA\radiant_quikqa
python -m radiant_gui.app
```

3.2 Main window layout

The Radiant QuikQA main window has three regions: a header bar, the analysis tab strip, and the per-tab result area.

Header bar

- Radiant logo and product name on the left.
- Active clinic dropdown. When a clinic is active, only that clinic's in-service machines appear in the machine dropdown, and the clinic's default chamber and electrometer settings auto-populate the TG-51 forms.
- Active machine dropdown. The selection here is the global machine context — every tab uses it. Switching machines refreshes each tab's per-machine default folders, tolerances, and beam catalog.

Outer tabs (grouped by tool family)

Starting in v0.8.2, related analyses are grouped behind a single outer tab with an inner notebook for the specific tool. v1.0.1 has nine outer tabs:

Outer tab	Inner tabs	Purpose
Winston-Lutz	—	Mechanical and radiation isocenter coincidence
MLC Tests	Picket Fence VMAT DRGS VMAT DRMLC	MLC leaf-position and dose-rate VMAT tests
Isocenter	—	Starshot variants (gantry / collimator / couch / Hancock 3D BETA)
Flat/Sym	—	Flatness and symmetry per Varian/Elekta/Siemens protocol
Imaging	Planar CT / CBCT	Phantom imaging — planar (19 phantoms) and CT/CBCT (BETA)
VMAT QA	Portal Dosimetry Log Files	Pre-treatment portal dosimetry and trajectory log analysis
TG-51	Photons Electrons	Reference dosimetry. Electrons tab includes a mode toggle for cross-cal (v1.0.1).

Outer tab	Inner tabs	Purpose
Anonymizer	—	DICOM anonymization for sharing test data
PDF Tools	—	Merge, split, watermark PDF reports

Result area (bottom of each tab)

- Status badge: PASS (green), ACTION (amber), FAIL (red), or UNKNOWN.
- One-line summary headline with the key metric and verdict.
- Metrics table showing the values the GUI considers most relevant for a quick scan. The full set of numbers is always in the PDF.
- Warnings list. Yellow warnings are informational (cal expires soon, beam commissioning aging, etc.); red warnings indicate something that should be addressed before signing the cal off (cal expired, baseline drift over 2.5%).
- "Open PDF report" button to open the generated report in the system PDF viewer.

3.3 Menu bar

Menu item	What it does
Setup → Clinics...	Manage clinic locations: clinic name, address, physicist credentials, in-service machines, default chamber + electrometer for TG-51, bookmarked folders.
Setup → Machines...	Manage the machine fleet: display names, Aria machine IDs, MLC types, all module tolerances, and the per-machine beam catalog (via the Beams... button on the right pane).
Setup → Preferences...	Per-workstation preferences: reports folder root, default tolerance preset, PDF stamping toggles, tolerance edit-lock.
Help → About	Version info and paths to the three JSON configuration files.

3.4 Workflow at a glance

- Pick the active clinic and active machine from the header dropdowns. One-time per session in most cases.
- Click the outer tab for the test family, then the inner tab for the specific tool.
- For PyLinac modules: select your input — a file, folder, or pair of files. For TG-51 modules: most fields auto-fill from clinic defaults and the beam catalog; you fill in the chamber readings and R50 / PDD.
- Adjust any per-run options (tolerances and baselines are pre-filled from the machine and beam catalog).
- Click the Run button. The analysis runs in a background thread; the result area updates with PASS/ACTION/FAIL and the headline metrics.
- Click "Open PDF report" to inspect the full output. The PDF is saved to the configured reports folder automatically.

4. Configuration System

Radiant QuikQA was designed for consulting workflows where one physicist serves multiple clinical sites. The configuration system has four layers: machines (the global fleet), clinics (the clinical sites), per-machine beam catalogs (new in v0.9.2), and per-workstation settings.

4.1 Machines

A Machine in Radiant QuikQA represents one physical linac. The fleet is global — a TrueBeam HD120 is one machine record even if it shows up at multiple clinics.

Open Setup → Machines... to see the current fleet. Each machine has:

Field	Description
Key	A short stable identifier (no spaces) — used internally and as the Aria machine match key.
Display name	Human-readable name shown in dropdowns and PDF reports.
Aria machine ID	The TreatmentMachineName as reported by Eclipse / Aria. The ESAPI launcher uses this to auto-select the profile.
MLC type	HD120, MLC120, or MLC120_IX. Drives PyLinac's MLC enum mapping in the Picket Fence module.
WL tolerances	Routine action/tolerance and SBRT action/tolerance in mm. SBRT values are typically tighter (0.5/0.75 mm vs 1.0/1.5 mm).
PF tolerances	Action and tolerance in mm; optional sag adjustment in mm.
VMAT tolerances	DRGS and DRMLC % deviation tolerances.
Starshot tolerance	Maximum allowed wobble diameter in mm.
FlatSym tolerances	Flatness and symmetry percentages.
CT tolerances	HU, scaling, thickness, low-contrast, CNR threshold.
PD tolerances	Gamma DTA / DD criteria and TG-218 pass-rate thresholds (v0.5+).
Default folders	Per-module default input folder. When you click Browse on that module's input row, the dialog opens here.
Beams (v0.9.2)	Per-machine beam catalog — see Section 4.4.

Tolerance edit lock

Tolerance fields in the Machines dialog are read-only by default. This is intentional — accidental clicks shouldn't be able to change clinical tolerances. To edit tolerances, open Setup → Preferences... and turn on "Allow editing of clinical tolerances". Toggle it back off after you're done.

4.2 Clinics (locations)

A Clinic represents a clinical site. Most installations only need the default "Default" clinic and never see the multi-site UI. For consulting practices serving multiple sites, clinics let you:

- Stamp clinic name, address, and physicist credentials on each PDF report from that site.
- Filter the machine dropdown to show only machines actually in service at the active clinic.
- Bookmark frequently-used input and output folders, scoped per clinic.
- Set a default machine for each clinic, pre-selected when you switch to that clinic.
- Set a default chamber and electrometer for TG-51 calibrations at this clinic (v0.9.2).

Open Setup → Clinics... to add or edit clinics. Each clinic has:

Field	Description
Key	Short stable identifier.
Display name	Shown in the active-clinic dropdown.
Clinic name + address	Optional, used for PDF stamping.
Physicist name + credentials	Optional, used for PDF stamping ("Todd Bernath, MS, DABR" style).
Reports folder	Optional override of the global reports folder root for this clinic.
Default machine	Pre-selected when this clinic is made active.
Machines in service	Subset of the global fleet that exists at this site. Drives the machine dropdown filter.
Default chamber (v0.9.2)	Chamber model, serial, N_Dw, and cal date. Auto-fills the TG-51 forms when this clinic is active. The cal date drives the calibration-expiry warning (v1.0.0).
Default electrometer (v0.9.2)	Electrometer model, serial, P_elec, and cal date. Same auto-fill and expiry-warning behavior as the chamber.
Folder bookmarks	Curated input and output folder shortcuts shown in dialogs at this site.

Auto-fill behavior: when you open a TG-51 tab, the chamber and electrometer fields populate from the active clinic's defaults. Fields you have already typed into are never overwritten — auto-fill only flows into blank fields. To pull defaults into a partially-filled form, clear the field and re-switch clinics.

4.3 Per-workstation preferences

Setup → Preferences... exposes settings that are specific to the current PC. These are stored in settings.json and don't sync between workstations — "where should reports go on this machine" is local to each machine.

Setting	What it controls
Active clinic at startup	Which clinic location is selected when the GUI opens.

Setting	What it controls
Default tolerance preset	"routine" or "sbrt" — affects which WL tolerance set is pre-checked when the WL tab opens.
Reports folder root	Top-level folder where PDF reports go. Per-machine subfolders are created automatically. Defaults to %LOCALAPPDATA%\RadiantQuikQA\Reports.
Allow tolerance editing	Unlocks tolerance fields in the Machines dialog. Off by default.
Include clinic name on reports	Stamping toggle.
Include physicist name on reports	Stamping toggle.
Include analysis date/time on reports	Stamping toggle.

4.4 Beam catalog (per machine, v0.9.2)

In Setup → Machines..., the right-hand pane has a Beams... button. Clicking it opens a dialog where you can configure per-machine beam energies, with each beam capturing:

Beam field	Used for
Energy label	"6 MV", "6 MV FFF", "6 MeV", etc. — appears in the TG-51 beam dropdown.
Modality	photon or electron — drives the beam dropdown filter on each TG-51 tab.
Last commission date	Drives the commissioning-age warning (v1.0.0): yellow >12 months, red >24 months.
Baseline value	%dd(10)_x for photons (e.g. 66.3), R50 in cm for electrons (e.g. 2.38). Auto-fills the form and drives drift warnings (v1.0.0).
Cone size (electrons)	Default electron applicator size (e.g. "10x10"). Auto-fills the form.
Lead foil required (photons)	Auto-checks the lead-foil box on the photon TG-51 tab.
Notes	Free-text per beam.

All beams commissioned on the same date

The BeamsDialog has a checkbox "All beams commissioned on the same date" at the top. When checked, the per-beam date field becomes a single shared date that applies to every beam in the catalog on save. This matches the common case where the whole machine is recommissioned in one session — no reason to type the date in multiple times.

Toggle off if you need different commission dates per beam (e.g. one energy was recommissioned later in isolation).

4.5 Per-machine default folders

Each analysis tab has a "Save as default" button next to the input path. Click it to remember the current folder as the per-machine default for that module. The next time you switch to this machine and open this tab, the Browse dialog will open in that folder.

Default folders are stored on the machine record (one field per module), so they follow the machine: switching from TrueBeam HD120 to Varian iX automatically retargets the Browse dialog to the iX's WL / PF / etc. folders.

Each tab also has a "Clear defaults" button that wipes saved default folders for the currently active machine on that tab only, after a confirmation.

5. Analysis Modules

Each PyLinac-backed module is a thin wrapper around a PyLinac analyzer. The wrappers add tolerance lookup against the machine profile, PASS/ACTION/FAIL classification, headline metrics extraction for the GUI, and branded PDF output. Numbers in each PDF come from PyLinac directly.

TG-51 modules (Section 6) use Radiant QuikQA's native engine; PyLinac is not involved.

5.1 Winston-Lutz

Module: WinstonLutz. PyLinac class: WinstonLutz. Use for MV EPID Winston-Lutz, kV/CBCT-derived Winston-Lutz, and BB-vs-CAX evaluation in general.

Inputs

- A folder of WL DICOM images (the standard Eclipse EPID export), OR a list of explicit file paths.
- PyLinac auto-detects kV/CBCT datasets and adjusts `low_density_bb` and `open_field` internally — you do not have to flag kV vs MV.

Per-run options

Option	Effect
BB size (mm)	Override the machine profile's default BB diameter. Default 5 mm.
SBRT tolerances	Check this box to use the machine's SRS/SBRT tolerance set instead of the routine TG-142 set. Affects only the verdict, not the numbers.
Open field	Force open-field mode. For large kV fields, PyLinac auto-handles this; manual toggle is for edge cases.
Low-density BB	BB is lower density than surroundings (e.g. plastic ball in metal jig). Auto-set by PyLinac for kV/CBCT.
Apply virtual shift	Apply the corrective shift to the BBs in the analysis. Useful for predicting what the numbers will be after a planned BB move.

PASS/ACTION/FAIL logic

- PASS: Max 2D CAX→BB distance $\leq$ action level.
- ACTION: action level $<$ Max 2D $\leq$ tolerance.
- FAIL: Max 2D $>$ tolerance.

5.2 Picket Fence

Module: PicketFence. PyLinac class: PicketFence. Use for MLC leaf-position accuracy. The machine's MLC type (HD120, Millennium 120, Millennium 120 on iX) is automatically mapped to PyLinac's MLC enum.

Inputs

- A single picket-fence DICOM image (EPID export).
- Optional: a Varian trajectory log file (.bin) for absolute (vs. fitted) reference. Strongly recommended on TrueBeams.

Per-run options

Option	Effect
Tolerance (mm)	Override the machine's PF tolerance. Defaults from machine profile (typically 0.5 mm per TG-142).
Action level (mm)	Override the machine's PF action level. Defaults from machine profile (typically 0.35 mm).
Sag adjustment (mm)	EPID sag correction. Defaults from machine profile (typically 0.0 unless characterized).
Crop (mm)	Pixels to crop from image edges before analysis. Default 3.

5.3 VMAT DRGS and VMAT DRMLC

Modules: VMAT-DRGS and VMAT-DRMLC. PyLinac classes: DRGS and DRMLC. Split across two inner tabs under MLC Tests to mirror clinical workflow — most physicists run them as two separate tests on different days.

DRGS (Dose-Rate / Gantry-Speed) tests the machine's ability to deliver variable dose rate during variable gantry speed. DRMLC (Dose-Rate / MLC-Speed) tests variable dose rate with variable MLC speed at constant gantry speed.

Inputs

Both tests require two images: an open field and a DMLC field. Provide them as either:

- A ZIP archive containing both, with PyLinac's conventional naming (filenames including "open" and "dmlc"), or
- Two file paths in the order [open, dmlc].

5.4 Starshot

Module: Starshot. PyLinac class: Starshot. Use for gantry, collimator, or couch rotation isocenter QA via a star pattern of exposed lines. Also hosts the 3D Isocenter (Hancock) BETA workflow.

Mode selector

- Gantry Starshot — gantry rotation wobble
- Collimator Starshot — collimator rotation wobble
- Couch Starshot — couch rotation wobble
- 3D Isocenter Evaluation (Hancock — BETA) — comprehensive 12+ image optimization

Hancock 3D Iso is BETA

The Hancock optimization has NOT been validated against RIT or vendor reference software. Use it in parallel with your existing 3D isocenter workflow until you have your own validation data. The GUI displays a yellow banner whenever the Hancock mode is selected, and the Run button prompts for confirmation before proceeding.

5.5 Flat/Sym

Module: FlatSym. PyLinac class: FieldAnalysis. Use for TG-142 monthly flatness and symmetry on open fields.

Per-run options

Option	Effect
Protocol	VARIAN, ELEKTA, or SIEMENS. Each vendor defines flatness and symmetry differently — pick the one your site policy specifies.
FFF beam	Check for flattening-filter-free beams. Switches PyLinac to inflection-derivative edge detection.
Flatness tolerance (%)	Override the machine's default. Typically 2.0%.
Symmetry tolerance (%)	Override the machine's default. Typically 2.0%.

5.6 Planar phantom imaging

Module: Planar. PyLinac module: planar_imaging. Supports 19 phantom types via a single dropdown. The wrapper accepts whichever kwargs each phantom class supports and silently drops the rest, so one entry point handles image-quality and alignment phantoms uniformly.

Supported phantoms include the Standard Imaging QC family (QC3, QcKV, FC2), Las Vegas variants (LasVegas, ElektaLasVegas), Leeds TOR (TOR, TORBlue), Doselab MC2/MC2kV/RLf, the Sun Nuclear family (SNCMV, SNCMV12510, SNCKV, SNCFSQA), PTW EPIDQC, IBA Primus A, IMTLRad, IsoAlign, and ACR Digital Mammography.

Inputs

- A single phantom image: DICOM, TIFF, PNG, or JPG.

5.7 CT / CBCT phantom QA (BETA)

Module: CT. PyLinac classes: CatPhan503/504/600/604/700, ACRCT, QuartDVT, TomoCheese.

BETA — validation required

The CT phantom QA module has NOT been validated against site-known reference values or vendor analysis software. Use it in parallel with your existing vendor analysis until you have side-by-side validation data on your own scanners and phantoms.

The GUI shows a yellow banner above all CT inputs. The Run button asks for confirmation before each run. All CT PDF reports carry a Mode: BETA metadata row.

5.8 Portal Dosimetry (v0.5+)

Outer tab: VMAT QA → Portal Dosimetry. Used for pre-treatment patient-specific QA using EPID-acquired dose images compared against the predicted dose from the TPS. Gamma analysis is performed per TG-218 conventions.

Inputs

- Predicted dose image (RTDose DICOM from the TPS / Eclipse Portal Dose calculation).
- Measured dose image (RTImage from the EPID-based portal dosimetry acquisition).

Per-run options

Option	Effect
DTA (mm)	Distance-to-agreement for gamma. Default from machine profile (typically 2.0 mm).
Dose difference (%)	Dose-difference criterion for gamma. Default from machine profile (typically 3.0%).
Threshold (%)	Low-dose threshold; pixels below this fraction of max are excluded from gamma. Default 10%.
Action pass rate (%)	Below this gamma pass rate, the analysis is flagged ACTION. Default 95%.
Tolerance pass rate (%)	Below this gamma pass rate, the analysis FAILS. Default 90%.

PASS/ACTION/FAIL logic

- PASS: gamma pass rate $\geq$ action pass rate.
- ACTION: tolerance pass rate $\leq$ gamma pass rate $<$ action pass rate.
- FAIL: gamma pass rate $<$ tolerance pass rate.

5.9 Log Files / VMAT trajectory log analysis

Outer tab: VMAT QA → Log Files. Analyzes Varian trajectory log files (.bin) for MLC leaf-position accuracy and gantry / collimator / dose-rate tracking across a delivered arc.

Inputs

- One or more .bin trajectory log files. The wrapper analyzes each independently and combines summary statistics.

Headline metrics

- Max / median / mean MLC leaf-position error across all logs.
- Gantry, collimator, couch tracking error.
- Dose-rate tracking error.
- Number of control points analyzed.

5.10 Anonymizer

Outer tab: Anonymizer. Strips patient-identifying tags from DICOM files for sharing test data with vendors, colleagues, or sharing internally without exposing PHI. Implements a defense-in-depth approach: known PHI tags are removed, free-text tags are regex-scrubbed, and UIDs are optionally regenerated.

Per-run options

Option	Effect
Regenerate UUIDs	On by default. Replaces StudyInstanceUID, SeriesInstanceUID, SOPInstanceUID with new ones to prevent cross-referencing.
Strip private tags	On by default. Removes vendor-private tags that may contain PHI.
Verify after write	On by default. Re-reads the output file to confirm no PHI fields survived.

5.11 PDF Tools

Outer tab: PDF Tools. Local utilities for working with the PDF reports Radiant QuikQA emits, plus arbitrary PDFs in your workflow:

- Merge — combine multiple PDFs into one (e.g. monthly QA bundle).
- Split — extract a page range from a PDF.
- Watermark — apply a text watermark (e.g. "DRAFT" or "PRELIMINARY") to every page.
- Add table of contents — auto-generate a bookmarked TOC from merged PDF page titles.

6. TG-51 Reference Dosimetry

The TG-51 outer tab houses Radiant QuikQA's native reference-dosimetry engines, accessible via two inner tabs: Photons and Electrons. The engines compute cGy/MU per AAPM TG-51 (Almond et al. 1999), the TG-51 Addendum (McEwen et al. 2014), and Report 374 (AAPM TG-51 Working Group 2022). All math is native; PyLinac is not involved on the TG-51 path.

The Electrons tab also hosts the v1.0.1 cross-cal mode, which implements TG-39 / TG-51 §VIII.A.2 parallel-plate cross-calibration.

6.1 Photon reference dosimetry

The Photons inner tab implements the full TG-51 photon protocol with two k_Q sources selectable per chamber:

- **ADDENDUM** — the 2014 Addendum's quadratic k_Q polynomial in $\%dd(10)_x$, with chamber-specific coefficients (A, B, C). This is the default for chambers where it is available.
- **LIBRARY** — TG-51 §11 substitution library, for chambers like the PTW 31010 Semiflex where the Addendum doesn't list a coefficient and the substitution path is the only option.

Form fields

Section	Fields
Header	Beam energy (auto-filled from beam catalog), measurement date, physicist.
Chamber	Model (dropdown, sorted by manufacturer), serial, N_{Dw} (Gy/C), cal date, k_Q source radio buttons.
Electrometer	Model (dropdown), serial, P_{elec} , cal date.
Geometry / environment / voltages	SSD or SAD radio, distance (cm), temperature (°C), pressure (kPa), relative humidity (%), $V_{operating}$, V_{high} , V_{low} , operating polarity radio.
Measurements	Five reading channels (M_{raw} , M_+ , M_- , M_{high} , M_{low}), each accepting up to 3 readings per channel. The form averages whatever is filled.
Beam quality / MU	MU delivered, $PDD(10)_x$ (auto-filled from beam catalog if configured), lead foil checkbox (auto-checked from beam catalog if configured).

Beam catalog integration

When the active machine has photon beams configured in its beam catalog, the beam-energy dropdown is driven by the catalog (not the static preset list). Selecting a beam from the dropdown auto-fills $PDD(10)_x$ and lead-foil-required from the BeamRecord. Manual override of any field is preserved on subsequent runs of the same session.

Headline output

- cGy/MU (the headline number, big and bold on the certificate page).
- All corrections: P_TP, P_ion, P_pol, P_elec, M_corrected.
- Beam quality: %dd(10)_x, k_Q, k_Q source citation.
- Dose: N_Dw, D_w at reference depth, MU delivered.
- Drift vs. baseline (if history exists for this machine + beam): %% deviation from the most recent calibration.
- Warnings (see Section 6.5).

6.2 Electron reference dosimetry

The Electrons inner tab implements TG-51 electron reference dosimetry using a two-reading P_gr_Q approach per AAPM Report 374. The k_Q decomposition is:

$$k_Q = P_{gr_Q} * k_{R50'} * k_{R50,ecal}$$

Where:

- P_gr_Q is the gradient correction at the reference depth d_ref — computed from a second M_raw reading taken with the chamber shifted by $0.5 \cdot r_{cav}$ from d_ref. For cylindrical chambers only; parallel-plate chambers use P_gr_Q = 1.0 by protocol convention.
- k_R50' is a chamber-independent empirical function of R50 (Burns/Ding/Rogers 1996):
 $k_{R50'} = 0.9905 + 0.0710 \cdot \exp(-R50/3.67)$.
- k_R50,ecal is a chamber-specific constant from TG-51 (1999) Table III (Table II for parallel-plate chambers).

Chamber catalog

The chamber dropdown is filtered to chambers with published TG-51 Table III data. The catalog ships with:

- Capintec PR06CG (k=0.900, r_cav=0.320 cm)
- Exradin A1 (k=0.915, r_cav=0.200 cm)
- Exradin A12 / A19 (k=0.906, r_cav=0.305 cm)
- NE 2561 (k=0.904, r_cav=0.370 cm — engine uses 0.350 cm for gradient correction per Table III footnote e)
- NE 2571 (k=0.903, r_cav=0.315 cm)
- PTW 30010 / 30011 / 30012 / 30013 (k=0.897 / 0.900 / 0.905 / 0.897, r_cav=0.305 cm)
- PTW 31013 (k=0.898, r_cav=0.275 cm)

PTW model numbers are the modern catalog designations; cross-walks to the historical N300xx numbers used in TG-51 Table III are documented in the catalog module's docstring.

Live d_ref display

The form computes $d_{ref} = 0.6 \cdot R50 - 0.1$ live as you type R50, and displays it in green next to the R50 entry. The engine uses the typed R50 to look up $k_{R50'}$ and to determine which depth to use for the gradient correction at $d_{ref} + 0.5 \cdot r_{cav}$.

Six reading channels

Unlike the photon tab's five channels, the electron tab has six. Channel six is M at ($d_{ref} + 0.5 \cdot r_{cav}$) — the second reading required for P_gr_Q. The form labels make the purpose explicit.

6.3 Cross-calibration of parallel-plate chambers (v1.0.1)

The Electrons tab has a calibration-mode radio at the top:

- Standard cal — single cylindrical chamber, single beam, the workflow described in Section 6.2.
- Cross-cal a parallel-plate chamber against a Farmer — the TG-39 / TG-51 §VIII.A.2 cross-cal procedure. New in v1.0.1.

When you switch to cross-cal mode:

- The chamber subframe is re-labeled "Cylindrical reference chamber".
- A new "Parallel-plate chamber" subframe appears, with a chamber dropdown listing PTW Roos (PTW 34001, $k=0.901$) and PTW Markus (PTW 23343, $k=0.905$). Both from TG-51 (1999) Table II.
- A new "Parallel-plate readings" subframe appears, with 5 reading channels (no shifted reading — PP chambers have a flat front face and no gradient correction).
- The Run button re-labels to "Run TG-39 cross-calibration".

Math

Both chambers are irradiated simultaneously (or sequentially with the same MU) at d_{ref} in the cross-cal beam. The engine computes the dose to water via the calibrated cylindrical chamber:

$$D_w = M_{cyl_corrected} * N_{Dw,cyl} * k_{Q,cyl}$$

And back-computes the effective N_{Dw} for the PP chamber:

$$N_{Dw,pp} = D_w / (M_{pp_corrected} * k_{R50'} * k_{R50,ecal,pp})$$

Note that $k_{R50'}$ is the same for both chambers because it depends only on R50, not chamber geometry. $P_{gr_Q,pp} = 1.0$ by protocol convention for parallel-plate chambers.

Cross-cal recommendations (TG-39)

Use the highest-energy electron beam available

TG-39 recommends performing the cross-cal in the highest clinically available electron beam ($R50 \geq 5$ cm, i.e. ≥ 16 MeV). The engine emits a yellow warning if $R50 < 5$ cm — the math still runs, but the resulting $N_{Dw,pp}$ is less robust for use at lower energies due to the smaller dynamic range of $k_{R50'}$.

Advanced Markus (PTW 34045) is intentionally not in the catalog yet. The standard Markus (PTW 23343) is the only Markus variant with its $k_{R50,ecal}$ value populated. If you have an Advanced Markus, contact me and I'll add it as a separate catalog entry.

Cross-cal record persistence

When "Save this calc to history" is checked, cross-cal results are stored under the PP chamber's serial number — not under a machine — because a cross-cal travels with the chamber. The same Roos #ABC123 used at two clinics shares a single cross-cal history. Storage location:

%LOCALAPPDATA%\RadiantQuikQA\tg51_cross_cal_history\

Filenames follow the pattern {pp_short_name}__{pp_serial}.json — e.g. PTW_Roos__ABC123.json.

6.4 Calibration history and drift trending

Every TG-51 calibration is appended to a per-(machine, beam) JSON history file when "Save this calc to history" is checked. The most recent record is used as the baseline for drift computation on the next cal at the same beam — no manual baseline entry required. To disable auto-baseline-pull, uncheck "Auto-pull baseline from history" on the form.

Storage locations:

History type	Path
Photon cal records	%LOCALAPPDATA%\RadiantQuikQA\tg51_history\
Electron cal records	%LOCALAPPDATA%\RadiantQuikQA\tg51_electron_history\
Cross-cal records	%LOCALAPPDATA%\RadiantQuikQA\tg51_cross_cal_history\

Each PDF report includes a trending plot on page 3 (when at least 2 records exist for the beam). The plot shows cGy/MU vs. measurement date, with $\pm 1\%$ and $\pm 2.5\%$ bands relative to the first record for visual scale.

6.5 Warning system (v1.0.0)

Each TG-51 calibration runs four families of post-calc checks after the dosimetry math completes. Warnings appear in the result panel's warnings list with severity prefixes ([YELLOW] or [RED]) and are also written to the PDF cert + worksheet pages. Warnings are informational only — they never block the Run.

Chamber and electrometer calibration expiry

- Computed from the cal_date stored on the clinic's default chamber / electrometer (or the cal date the user typed into the form). The 24-month expiration is exactly 24 months from cal_date (730 days, computed via 30.4375 days/month approximation).
- RED warning: cal_date + 24 months is past the measurement date. The cal is no longer traceable; the chamber or electrometer should be sent to ADCL.
- YELLOW warning: ≤ 3 months until expiry. Schedule recalibration.
- Silent: more than 3 months until expiry.
- Silent: cal_date not recorded (we don't nag about empty fields).

Beam commissioning age

- Computed from BeamRecord.last_commission_date to the cal's measurement date — not from "now". This means the warning reflects the beam's commissioning age at the time of THIS cal, important for audit trail when reviewing a year-old report.
- YELLOW warning: $12 < \text{commission_age} \leq 24$ months.
- RED warning: commission_age > 24 months. Consider recommissioning before trusting the baseline comparison.
- Silent: commission_date not configured, or commission_age ≤ 12 months.

Baseline drift vs. BeamRecord.baseline_value

- Photons: measured PDD(10)_x vs. catalog. YELLOW > 1.0% absolute, RED > 2.5% absolute (TG-142 thresholds).
- Electrons: measured R50 vs. catalog. YELLOW > 1.0 mm absolute, RED > 2.5 mm absolute (TG-142 thresholds).
- Silent: no baseline configured (baseline_value = 0.0 in the BeamRecord).

Cal-age display on PDF reports

Regardless of warning state, every TG-51 PDF cert page shows two new rows in the Equipment block (v1.0.1):

Chamber cal: 2024-09-15 (8.0 months old, within 2-year cycle)

Electrometer cal: 2024-09-15 (8.0 months old, within 2-year cycle)

Expired cals are stamped EXPIRED with the number of months past:

Chamber cal: 2023-05-15 (24.5 months old, EXPIRED (0.5 months past))

This ensures the audit trail in every report is unambiguous, regardless of whether the cal was within or past the 2-year cycle when run.

7. PDF Reports

Every successful analysis publishes a branded PDF report. PyLinac modules use PyLinac's `publish_pdf` method; TG-51 modules use Radiant QuikQA's native renderer. The reports are designed to drop straight into a clinical QA binder or chart-check workflow.

7.1 Anatomy of a PyLinac report

- Header: Radiant Medical Physics logo at the top of the first page.
- Metadata box: machine name, test parameters, status (PASS/ACTION/FAIL), and for BETA modules a "Mode: BETA" row.
- Results section: PyLinac's native numerical results — every metric for the module, not just the GUI headlines.
- Figures: PyLinac's visualizations (e.g. WL CAX-vs-BB scatter, PF leaf-error plot, CT slice montages with overlay ROIs).

7.2 Anatomy of a TG-51 report

TG-51 reports are 3 pages:

10. Certificate: machine, beam, date, physicist, chamber + electrometer details with cal dates and cal age, all corrections (P_TP, P_ion, P_pol, P_elec, M_corrected, k_Q), beam quality (PDD or R50, k_Q source), dose (N_Dw, D_w), and the headline cGy/MU number in large bold red.
11. Worksheet: every input value, the full formula chain (P_TP, P_ion, P_pol equations and substituted values), all working numbers, the full warnings list, and the full k_Q source citation. Designed for physicist signoff and audit.
12. Trending: cGy/MU vs. measurement date for this (machine, beam), with $\pm 1\%$ and $\pm 2.5\%$ bands relative to the first record. Omitted if fewer than 2 history records exist.

7.3 Where reports are saved

By default, into per-machine subfolders under the reports folder root configured in Preferences:

`%LOCALAPPDATA%\RadiantQuikQA\Reports\<machine_key>\`

Each module suggests a filename based on the module and date — for example, WL_2026-05-15.pdf, CT_CatPhan504_2026-05-15.pdf, TG51_TrueBeam_HD120_6MV_20260515_142233.pdf.

The user can override this at the Output PDF row on each tab; Browse → Save as default remembers the destination folder per-machine.

7.4 PDF stamping toggles

Three optional stamps are controlled in Setup → Preferences:

- Clinic name — from the active Location.
- Physicist name — "Name, Credentials" from the active Location.
- Analysis date and time — when the report was generated.

8. Command-Line Interface

Every analysis module is also exposed as a CLI subcommand:

```
RadiantQuikQA.exe <module> [options]
```

Or, when running from source:

```
python -m radiant_core.cli <module> [options]
```

The CLI is what the bundled Eclipse ESAPI launcher calls. It can also be used standalone for scripted workflows, scheduled tasks, or unattended batch processing.

8.1 Global options

Option	Effect
--json	Emit machine-readable JSON to stdout instead of human-readable text. Used by the ESAPI launcher to parse results back into Eclipse.
-h, --help	Help text for the top-level command or any subcommand.

8.2 Exit codes

Code	Meaning
0	PASS — analysis succeeded and is within tolerance.
1	ACTION — analysis succeeded but exceeded the action level (still within tolerance).
2	FAIL — analysis succeeded but exceeded tolerance.
3	Analysis error — an exception was raised during the run.

8.3 Module subcommands

Winston-Lutz

```
RadiantQuikQA.exe wl --input <folder> --output <pdf> --machine <key> [--sbprt]
```

Picket Fence

```
RadiantQuikQA.exe pf --input <dcim> [--log <bin>] --output <pdf> --machine <key>
```

VMAT

```
RadiantQuikQA.exe vmat --test DRGS|DRMLC --input <zip> --output <pdf> --machine <key>
```

CT (BETA)

```
RadiantQuikQA.exe ct --input <folder> --output <pdf> --machine <key> --phantom CatPhan604
```

Portal Dosimetry

```
RadiantQuikQA.exe pd --predicted <rtdose> --measured <rtimage> --output <pdf> --machine <key>
```

TG-51 photons (v0.8.0+)

```
RadiantQuikQA.exe tg51 --input <json> --output <json> [--pdf-output <pdf>] [--save-history] [--use-history-baseline]
```

The --input JSON describes the calibration; its schema matches the TG51PhotonInput dataclass. The --output JSON is the full TG51PhotonResult. With --pdf-output the 3-page PDF is also rendered. --save-history appends to the per-(machine, beam) history file.

TG-51 electrons (v1.0.1)

```
RadiantQuikQA.exe tg51-e --input <json> --output <json> [--pdf-output <pdf>] [--save-history] [--use-history-baseline]
```

Same shape as tg51, with the electron schema. Schema matches TG51ElectronInput.

TG-51 / TG-39 cross-cal (v1.0.1)

```
RadiantQuikQA.exe tg51-cross-cal --input <json> --output <json> [--save-history]
```

Schema matches CrossCalInput; output is CrossCalResult. With --save-history the result persists under the PP chamber's serial number.

Other subcommands

- anonymize — strip PHI from a folder of DICOMs.
- merge-pdfs — combine multiple PDFs.
- Starshot, FlatSym, and Planar are currently GUI-only. They will be added to the CLI in a future release.

9. Eclipse ESAPI Integration

Radiant QuikQA ships with a bundled C# launcher that registers as an Eclipse v15.6 published script. The launcher provides a one-click path from inside Eclipse to the Radiant QuikQA CLI.

9.1 How it works

13. A physicist opens Eclipse, opens a patient with the relevant treatment machine, and runs the Radiant QuikQA published script.
14. The launcher reads the TreatmentMachineName via ESAPI and maps it to a Radiant QuikQA machine key (by Aria machine ID).
15. The launcher invokes RadiantQuikQA.exe with the appropriate subcommand, the machine key, and an input folder typically derived from the EPID export path.
16. RadiantQuikQA runs the analysis, writes the PDF, and exits with a status code.
17. The launcher parses the JSON output (when invoked with --json) and surfaces the PASS/ACTION/FAIL verdict back to the Eclipse user.

9.2 Why the launcher uses C# 5

Eclipse v15.6's ESAPI runtime is constrained to C# 5: no string interpolation, no null-conditional operators, no pattern matching. The launcher's .csproj is pinned with:

- TargetFrameworkRootPath = .NET 4.8
- PlatformTarget = x64 (matches Eclipse's 64-bit host)
- LangVersion = 5 (forces C# 5 compatibility)

Do not modify these settings in the .csproj. They are intentional and have been validated against the Eclipse 15.6 ESAPI runtime.

9.3 Where the published script lives

The bundled build_all.ps1 pipeline publishes the launcher to:

```
\\dp-varimg01\va_data3$\ProgramData\VISION\PublishedScripts\
```

Eclipse picks it up from there at next launch. No per-workstation install is needed for users — only the workstation running the analysis needs RadiantQuikQA.exe installed.

9.4 UI-thread safety

All ESAPI data access in the launcher runs on the Eclipse UI thread, per Varian's ESAPI design. The launcher reads what it needs synchronously, then spawns RadiantQuikQA.exe as a separate process. The Python side is fully decoupled — Radiant QuikQA itself has no ESAPI dependency.

10. DICOM Inspector

Radiant QuikQA includes a DICOM inspection utility that reads RT-image geometry tags from a folder of DICOM files and reports what the selection looks like clinically. It is most useful for Starshot, where the physicist needs to know which axis is varying in their image selection.

10.1 What it reports

- Number of files scanned and number with usable RT-image tags.
- Which mechanical axis is varying — gantry, collimator, or couch — based on which set of angles has more than one distinct value.
- The distinct angle values (degrees) for each axis.
- The machine name reported in the DICOM header (TreatmentMachineName).
- Modality and RT image label.
- Files that could not be read (with reasons).

10.2 Angle equivalence threshold

Two angles within 0.5° of each other are treated as identical. Varian EPID exports sometimes write 90.00° vs 89.998° for the same physical setting; treating those as different would produce false "axis is varying" reports.

10.3 Starshot minimum-images recommendation

PyLinac will run a starshot with as few as 3 images, but its fit confidence is much higher with 4 or more. The inspector reports a recommendation if you have fewer than 4 distinct angles on the varying axis.

11. Maintenance and Updates

11.1 Updating Radiant QuikQA

Updates come in three flavors:

- Patch zips (e.g. v1.0.1, v1.0.2) — contain only the files that changed since the previous release. Extract over your existing install at C:\Radiant_QuikQA\radiant_quikqa\.
- Full source zips (major releases) — replace the entire radiant_quikqa folder.
- Pre-built distribution zips — the dist/ output of build_all.ps1, suitable for handoff to a colleague who doesn't need to develop on the source.

Either way, configuration data (machines, locations, settings, beam catalogs, TG-51 history) is unaffected — those live in %LOCALAPPDATA%\RadiantQuikQA\ and survive updates.

11.2 Backing up configuration

To back up your complete configuration:

```
$src = "$env:LOCALAPPDATA\RadiantQuikQA"
$dst = "$env:USERPROFILE\Backups\RadiantQuikQA_$(Get-Date -Format yyyyMMdd)"
Copy-Item -Path $src -Destination $dst -Recurse
```

11.3 Resetting to defaults

To wipe configuration and bootstrap defaults on next launch, delete or rename the JSON files (the reports folder is independent):

```
Remove-Item "$env:LOCALAPPDATA\RadiantQuikQA\machines.json"
Remove-Item "$env:LOCALAPPDATA\RadiantQuikQA\locations.json"
Remove-Item "$env:LOCALAPPDATA\RadiantQuikQA\settings.json"
```

On next launch, Radiant QuikQA will bootstrap the default fleet (TrueBeam HD120, TrueBeam Millennium, Varian iX) and a single "Default" clinic.

11.4 Tests

A pytest test suite is included for sanity-checking module wrappers, configuration round-trips, and the TG-51 engines:

```
cd C:\Radiant_QuikQA\radiant_quikqa
python -m pytest -q
```

v1.0.1 ships with a test suite covering: Winston-Lutz, Picket Fence, VMAT, Starshot, CT, Portal Dosimetry, the DICOM inspector, store round-tripping, and the TG-51 photon and electron engines (38 passing tests for the electron engine alone, pinned to Burns/Ding/Rogers k_R50' values verified against the upload spreadsheet to 0.05% agreement).

11.5 Sharing the application with a colleague

To package Radiant QuikQA for handoff to a colleague:

18. On your workstation, do a clean rebuild:

```
cd C:\Radiant_QuikQA\radiant_quikqa
Remove-Item -Recurse -Force build, dist -EA SilentlyContinue
.\build_all.ps1 -VarianApiPath "C:\esapi_refs\"
```

19. Zip the entire dist\ folder produced by PyInstaller. The zip is typically 200–400 MB.
20. Send the zip. Your colleague extracts it to any folder of their choosing and runs RadiantQuikQA.exe directly. No Python install needed.
21. On first launch, configuration bootstraps under their workstation's %LOCALAPPDATA%\RadiantQuikQA\. They walk through Setup → Clinics and Setup → Machines to set up their own fleet.

If your colleague's Norton flags the .exe

PyInstaller-packaged Python binaries are sometimes flagged heuristically by endpoint protection. Have them add the install folder to Norton's exclusion list. Exclude the entire folder, not individual files — the .exe extracts to a temp dir on launch, and a per-file exclusion misses the extracted files.

12. Troubleshooting

12.1 PDF write fails with PermissionError

Cause: the previous copy of the PDF is still open in a viewer (Adobe, Edge, etc.). Windows holds an exclusive lock.

Fix: close the open PDF and run again. The GUI catches this case specifically and shows the friendlier message: "Close any open copy of the report and run again."

12.2 Phantom not detected (planar)

Cause: usually the wrong phantom selected in the dropdown, or the phantom not centered on CAX in the image.

Fix: confirm the dropdown phantom matches the physical phantom imaged. Confirm the phantom is centered on CAX. If the image polarity is inverted, try the "Invert image polarity" checkbox.

12.3 CT scan extent does not match module configuration

Cause: the selected phantom class expects more modules than your scan covers in Z.

Fix: verify the phantom class matches the physical phantom (CatPhan504 has 3 modules; 604 has 4 modules). If correct, re-scan with a longer scan length.

12.4 TG-51 cGy/MU disagrees with my spreadsheet by ~0.1%

Expected on the first electron cal you run. Two reasons:

- The engine uses the full TG-51 two-reading $P_{gr,Q}$ approach (M at d_{ref} AND M at $d_{ref} + 0.5 \cdot r_{cav}$). Some clinic spreadsheets use the simplified single-reading + EPOM shift approach, which Report 374 (2022) notes is ~0.1% less accurate.
- PTW 30013 specifically: the engine uses $k_{R50,ecal} = 0.897$ (verified TG-51 Table III value). Some spreadsheets used 0.898 from a historical transcription confusion with the N31003 row. The 0.1% difference here compounds with the $P_{gr,Q}$ approach above.

Combined, you may see 0.1–0.2% disagreement on the first electron cal. Anything larger than 0.5% should be investigated as a potential engine bug or chamber data issue.

12.5 TG-51 form shows "(no published $k_{R50,ecal}$)"

Cause: you picked a chamber whose row in the catalog is marked NOT_POPULATED (no TG-51 Table III value). The engine refuses to compute k_Q from extrapolated values.

Fix: pick a chamber with published Table III data. For electrons, the workhorse Farmer chambers (NE2571, Exradin A12, PTW 30010–30013) all have published values. For IBA FC65G or similar chambers not in Table III, the recommended workflow is to cross-calibrate them in a photon beam against a chamber that IS in Table III (or contact Todd to add an extrapolated entry to the catalog if the literature supports one).

12.6 Cross-cal N_Dw,pp is outside the typical 5e7 to 1e9 Gy/C range

Cause: very likely a units or sign error in the readings. The cross-cal engine emits a warning when N_Dw,pp falls outside the typical clinical range.

Fix: verify all readings are in nC (not C, not pC). Verify M+ and M- have the correct signs per your electrometer's convention. Verify the cyl N_Dw is also in Gy/C (not Gy/nC). Verify R50 is in cm.

12.7 Norton AV blocks the executable

Cause: PyInstaller-packaged Python binaries are sometimes flagged heuristically by endpoint protection.

Fix: add the install folder to Norton's exclusion list. Exclude the entire C:\Radiant_QuikQA\ folder, not individual files.

12.8 Save dialogs default to OneDrive Documents

Cause: Documents has been redirected to OneDrive on your workstation. Radiant QuikQA itself uses %LOCALAPPDATA% for its config and reports, but Windows file-save dialogs may still default to the OneDrive Documents path the first time they open.

Fix: browse once to your preferred output location and click Save as default on that tab. After that, the per-machine default folder system takes over.

12.9 ESAPI launcher fails to compile

Cause: missing Varian API references at the path expected by the .csproj.

Fix: pass the correct path via the -VarianApiPath flag to build_all.ps1:

```
.\build_all.ps1 -VarianApiPath "C:\esapi_refs\"
```

The path should contain VMS.TPS.Common.Model.API.dll and its companions. Do not modify the .csproj's TargetFrameworkRootPath, PlatformTarget, or LangVersion settings.

12.10 Tests fail after upgrading PyLinac

Cause: PyLinac's API or result schema changed between minor versions.

Fix: roll PyLinac back to the pinned version (3.43.2 for v1.0.1). The Radiant QuikQA wrappers are validated against the pinned version and are not guaranteed against newer or older PyLinac releases.

13. Appendix

13.1 Default tolerances by module

Module / Threshold	Default
Winston-Lutz (routine)	1.0 / 1.5 mm
Winston-Lutz (SBRT)	0.5 / 0.75 mm (HD120) or 0.75 / 1.0 mm
Picket Fence	0.35 / 0.5 mm
VMAT DRGS	1.5%
VMAT DRMLC	1.5%
Starshot	1.0 mm
Flatness	2.0%
Symmetry	2.0%
CT HU	40 HU
CT scaling	1.0 mm
CT thickness	0.2 mm
CT low-contrast	1.0 mm
CT CNR	15
Portal Dosimetry DTA / DD	2.0 mm / 3.0%
Portal Dosimetry pass rates	95% (action) / 90% (tolerance)
TG-51 cal expiry	24 months exact
TG-51 cal expiry warning lead	3 months
TG-51 commission age (yellow / red)	12 / 24 months
TG-51 PDD drift (yellow / red)	1.0% / 2.5% absolute
TG-51 R50 drift (yellow / red)	1.0 mm / 2.5 mm absolute

13.2 File locations reference

Item	Path
Install root	C:\Radiant_QuikQA\

Item	Path
Source tree	C:\Radiant_QuikQA\radiant_quikqa\
Eclipse launcher	C:\Radiant_QuikQA\radiant_quikqa\eclipse_launcher\
Published scripts share	\\dp-varimg01\va_data3\$\ProgramData\VISION\PublishedScripts\
Config root	%LOCALAPPDATA%\RadiantQuikQA\
Default reports root	%LOCALAPPDATA%\RadiantQuikQA\Reports\
Settings file	%LOCALAPPDATA%\RadiantQuikQA\settings.json
Machines file	%LOCALAPPDATA%\RadiantQuikQA\machines.json
Locations file	%LOCALAPPDATA%\RadiantQuikQA\locations.json
TG-51 photon history	%LOCALAPPDATA%\RadiantQuikQA\tg51_history\
TG-51 electron history	%LOCALAPPDATA%\RadiantQuikQA\tg51_electron_history\
TG-51 cross-cal history	%LOCALAPPDATA%\RadiantQuikQA\tg51_cross_cal_history\

13.3 Version history

v1.0.1 (current):

- TG-39 / TG-51 §VIII.A.2 parallel-plate chamber cross-calibration. PP catalog ships with PTW Roos (k=0.901) and PTW Markus (k=0.905); engine + persistence + GUI mode toggle on the Electrons tab.
- CLI subcommands: tg51-e and tg51-cross-cal.
- Photon PDF cert page now shows chamber + electrometer cal date and cal age.

v1.0.0:

- Photon TG-51 tab gains beam-catalog-driven dropdown (mirrors v0.9.2 electron behavior).
- Four-family warning system: chamber + electrometer cal expiry (24-month cycle, 3-month warning lead, RED on expired); beam commissioning age (yellow >12mo, red >24mo); PDD drift (1% yellow, 2.5% red); R50 drift (1 mm yellow, 2.5 mm red).
- Electron PDF cert page shows chamber + electrometer cal date and cal age.

v0.9.2:

- BeamRecord dataclass added to MachineRecord. Per-machine beam catalog dialog (BeamsDialog) with photon and electron beams, baseline values, last-commission dates, cone size, lead-foil flag.
- "All beams commissioned on the same date" checkbox for bulk date entry.
- LocationRecord gains 8 fields for default chamber + electrometer settings.
- TG-51 tabs auto-fill chamber and electrometer from active clinic; beam dropdown driven by configured beams; R50 / cone size auto-fill on beam selection.
- Fully backward compatible — old JSON files load with defaulted new fields.

v0.9.0:

- TG-51 electron reference dosimetry. Outer tab restructure: TG-51 Photons → TG-51 with inner Photons / Electrons tabs.
- 11 chambers in the electron catalog with verified TG-51 Table III data (Capintec, Exradin A1/A12/A19, NE2561/2571, PTW 30010/30011/30012/30013/31013), plus 8 NOT_POPULATED entries for chambers without published Table III values.
- Two-reading P_{gr}Q approach per AAPM Report 374.
- Per-(machine, beam) electron history; 3-page PDF report.

v0.8.x:

- TG-51 photon reference dosimetry with 2014 Addendum and LIBRARY k_Q sources.
- 3-readings-per-channel averaging on the form.
- MLC Tests / Imaging / VMAT QA inner-notebook tab restructure.

v0.4.0 → v0.7.x:

- CT/CBCT phantom QA (BETA).
- Portal Dosimetry.
- Log Files / trajectory log analysis.
- Anonymizer, PDF Tools.
- Multi-clinic config; per-machine default folders; DICOM inspector.

13.4 Acknowledgments

Radiant QuikQA's PyLinac-backed modules are built on PyLinac (<https://pylinac.readthedocs.io>), an open-source medical physics analysis library. Numbers in those reports come from PyLinac. Radiant QuikQA owns the clinical workflow, tolerance dispatch, configuration system, multi-clinic UX, ESAPI integration, and the native TG-51 reference dosimetry engines (photon, electron, and TG-39 cross-cal).

TG-51 engine math references:

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